Institut Teknologi Bandung

Directorate of Continuing Professional Education

in partnership with

CHAPTER 1

What Is Deep Learning?

Putting AI, machine learning, and deep learning in their right places – then separating what the field has actually achieved from what it has merely promised.

Duration 90 minutes

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Source Chollet & Watson, Deep Learning with Python 3e – chapter 1

LEARNING OUTCOMES

Session Objectives

By the end of this session, participants can:

1. Place **AI, machine learning, and deep learning** in the correct relationship – which contains which, and since when.
2. Explain the **inversion**: classical programming outputs answers, machine learning outputs rules.
3. Name the **three things** every machine learning system needs, and say what breaks when one of them is missing.
4. Describe how deep learning works through **weights, a loss function, and backpropagation** – without deriving a single equation.
5. Separate **what deep learning has demonstrably done** from **what is still a claim**, and name what triggered the two previous AI winters.

BOOK
Chapter 1 — full text

NOTEBOOK
01_the_ml_paradigm.ipynb

ROADMAP

What this chapter is for

Chapter 1 teaches no code at all. Its job is more basic: to make sure everyone in the room uses the same words for the same things.

Definitions and history

Three nested circles – AI, ML, DL – plus one branch that is **not** machine learning at all.

1.1 – 1.5

How it works

Representations, weights, loss, backpropagation. Chollet explains it with **three figures**, not equations.

1.6 – 1.7

Achievement vs hype

What deep learning has **already** done, and why the field went cold twice before.

1.8 – 1.12

01

AI, machine learning, deep learning

Three terms the press uses interchangeably. They are nested.

Three nested circles, not three synonyms

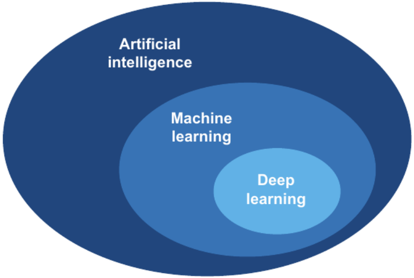
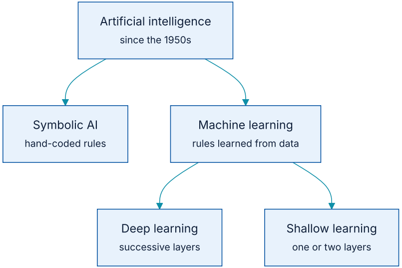


Figure 1.1 — deep learning is a subfield of machine learning, which is a subfield of AI. — Chollet & Watson, *Deep Learning with Python*, 3rd ed. (Manning)

The nesting is the whole point: every deep learning system is a machine learning system, and every machine learning system is an AI system – **but not the other way round**

SECTION 1.2

...and one branch that sits outside machine learning



Symbolic AI is inside AI but outside machine learning. Shallow learning is inside machine learning but outside deep learning.

Symbolic AI: what it was, and where it stopped

Artificial intelligence was born in the 1950s. The working definition: *the effort to automate intellectual tasks normally performed by humans*.

For roughly thirty years the dominant approach was **symbolic AI**: programmers wrote the rules by hand, and the machine executed them.

Where it worked

Well-defined, logical problems. Chess is the canonical win.

Where it collapsed

Fuzzy problems – recognising an image, understanding speech. Nobody can write down those rules, because nobody is aware of using them.

> Machine learning rose from the 1990s onward precisely because **the fuzzy problems were the ones left over**

SECTION 1.2

The question that opened the field, in 1843

> *The Analytical Engine has no pretensions whatever to originate anything. It can do whatever we know how to order it to perform.*
>
> Ada Lovelace, 1843 — on Charles Babbage's machine

Lovelace's objection is the oldest question in the field, and it is not rhetorical: **can a machine ever produce something we did not put into it?**

…and the claim that answered it, in 1956

> *Every aspect of learning or any other feature of intelligence can in principle be so precisely described that a machine can be made to simulate it.*
>
> John McCarthy et al., the Dartmouth workshop proposal, 1956

These two statements are 113 years apart and they disagree. Machine learning is a **partial** answer to Lovelace: the machine does produce something we never wrote down – **the rules**

02

The inversion

What used to be the output becomes the input.

SECTION 1.3

Machine learning runs programming backwards

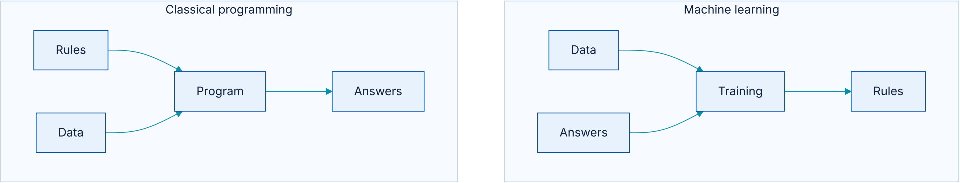


Figure 1.2 — classical programming is fed rules and data and emits answers; machine learning is fed data and answers and emits rules.

SECTION 1.3

The consequence people underestimate

> A machine learning system is **trained, not programmed**. If you have no worked examples, there is **nothing to train**, no matter how good the architecture is.

This is why the hardest part of most projects turns out to be the dataset rather than the model. Chapter 6 is devoted to that problem.

Three things every ML system needs

1 · Input data points
Sound files for speech recognition; photographs for image tagging. This is what gets transformed.

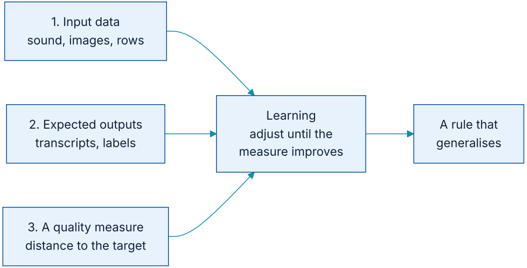
2 · Examples of expected output
Human transcripts for the sound files; tags such as *dog* or *cat* for the pictures.

3 · A way to measure quality
The distance between what the algorithm currently says and what it should say. This is the **feedback signal**.

That adjustment step, driven by the feedback signal, is **exactly what the word *learning* means here**

SECTION 1.4

The three, drawn as one pipeline



All three feed the same adjustment step. Remove any one of them and the arrow into it disappears.

SECTION 1.4

Lose one, and it stops being machine learning

MISSING INGREDIENT	WHAT YOU ACTUALLY HAVE	WHAT FOLLOWS
Expected outputs	Raw data with no labels	Not supervised learning. Either label it, or move to unsupervised or self-supervised methods.
A quality measure	Data and labels, no agreed metric	The model can be trained but **not judged** — and this is the most common way real projects fail.
Representative inputs	Examples unlike production conditions	It works on the laptop and fails in the field. Chapters 5 and 6 give this its proper names.

Representation: the idea that runs through the whole book

> Same picture, same task difficulty changed entirely. So machine learning is **the search for a representation that makes the task easy**

- *Select all red pixels* — easy in **RGB**.
- *Make the image less saturated* — easy in **HSV**.

The worked example: change the axes, and the rule appears

1. **Raw data.** White points and black points scattered in an (x, y) plane. No simple rule separates them.
2. **Coordinate change.** Move the origin, rotate the axes. Nothing about the data changed – only how it is written down.
3. **Better representation.** The rule is now one sentence: *black points are those with x > 0*.

> No model got smarter. Only the representation changed – and that is the work that used to be called **feature engineering** and was done by hand.

Why hand-written rules do not scale

Possible — up to a point

Rules such as *count the closed loops*, or vertical and horizontal pixel histograms, do a decent job on handwritten digits.

But brittle, and miserable to maintain

Every new handwriting sample that breaks your carefully reasoned rules forces a new transformation and a new rule — **and you must check it against every rule you already wrote**.

Hypothesis space: what the algorithm is allowed to search

> In the two-dimensional example, the hypothesis space was **the space of all possible coordinate changes**

Choosing a model architecture, in later chapters, is exactly the act of choosing a hypothesis space.

SECTION 1.4

The whole field, in one sentence

> *Machine learning is searching for useful representations and rules over some input data, within a predefined space of possibilities, using guidance from a feedback signal.*
>
> Chollet & Watson, section 1.4

That one idea covers a startling range of tasks – from autonomous driving to answering questions in natural language.

03

The "deep" in deep learning

Not deeper understanding. Just more layers.

SECTION 1.5

"Deep" is a statement about architecture, nothing more

> *The "deep" in "deep learning" isn't a reference to any kind of deeper understanding achieved by the approach; rather, it stands for this idea of successive layers of representations.*
>
> Chollet & Watson, section 1.5

The number of layers is the model's **depth**. Modern networks run to tens or hundreds of them.

SECTION 1.5

Each layer moves the data one step closer to the answer

Figures 1.5–1.6 — a four-layer network for digit classification. Intermediate representations get further from the pixels and closer to the answer.

Chollet calls this **information distillation**: irrelevant information is filtered out, relevant information is amplified. All the layers are learned **at once, automatically**, from the training data.

Two better names the field never adopted

> Its opposite has a name too: **shallow learning**, which learns only one or two layers of representation – for instance a pixel histogram followed by a classification rule.

- *layered representations learning*
- *hierarchical representations learning*

Either would have been more accurate. **Deep** won on history, not on precision.

04

How deep learning actually works

Three figures, and not one equation.

Step 1 — weights parameterise what a layer does

> Learning, then, means **finding values for every weight** such that the network maps its example inputs to their correct targets. A large network has tens of millions of them.

Step 2 — the loss function measures how far off you are

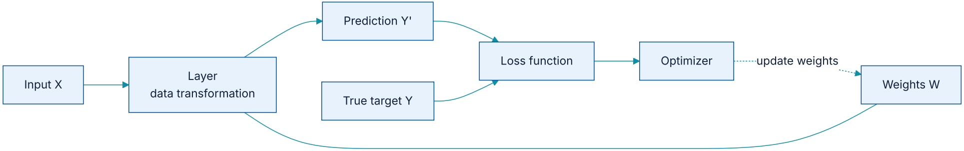
> That single number is the entire feedback signal. **Everything the network learns, it learns from that one score.**

Step 3 — backpropagation turns the score into an adjustment

The weights start out **random**, so the first outputs are nonsense and the first loss is high. What makes it work is **repetition of the loop**, not a lucky starting point.

SECTION 1.6

The three figures, joined into one loop



Figures 1.7–1.9 combined: weights parameterise the layer, the loss scores the prediction against the target, and the optimizer uses that score to move the weights.

This is called the **training loop**, repeated over thousands of examples. Chapter 2 takes every box in this picture apart.

The same loop, small enough to read in one sitting

We generate data from a law we choose – `y = 2x + 1` – and then **throw the law away**. The program never sees it. The question is whether it can recover it.

setup — data from a law we then hide

PYTHON

```
import numpy as np

rng = np.random.default_rng(0)
X = rng.uniform(-1, 1, size=(200, 1))
Y = 2 * X + 1 + rng.normal(0, 0.05, size=(200, 1))   # the law, plus a little noise

# The weights start random -- exactly as the chapter says.
W, b = rng.normal(size=(1, 1)), np.zeros((1,))
print(f"before training:  W {W[0, 0]:+.3f}   b {b[0]:+.3f}")
```

OUTPUT

```
before training:  W +0.126   b +0.000
```

Four lines that are the three figures

the training loop itself

PYTHON

```python
for step in range(600):
    Y_pred = X @ W + b                  # the layer: a data transformation (fig 1.7)
    loss = np.mean((Y_pred - Y) ** 2)   # the loss: how far off are we? (fig 1.8)

    grad = 2.0 * (Y_pred - Y) / len(X)  # gradient of the loss (fig 1.9)
    W -= 0.5 * (X.T @ grad)             # step downhill
    b -= 0.5 * grad.sum()

    if step % 200 == 0:
        print(f"step {step:3d}  loss {loss:.5f}  W {W[0, 0]:+.3f}  b {b[0]:+.3f}")
```

Forward pass, loss, gradient, update. **Every training loop in this book is that shape**, however large the model gets.

What comes out is the rule itself

OUTPUT

```
step   0  loss 4.02891  W +0.126  b +0.000
step 200  loss 0.00932  W +1.842  b +0.968
step 400  loss 0.00268  W +1.972  b +0.997
done      loss 0.00251  W +1.994  b +1.000
```

> The weights travel from random to **$W \approx 2$, $b \approx 1$** – the law that generated the data, which we never wrote down. That is *rules as output* from figure 1.2, made concrete.

- Not one line told the program the answer was 2 and 1.
- It was given only **data, answers, and a measure of loss** — the three required ingredients.
- Chapter 2 swaps this NumPy for tensors and automatic differentiation. The shape of the loop **does not change**.

05

Why deep learning won

Three properties, and one change that left the alternatives behind.

Property 1 — simplicity

Deep learning learns **all** the features in one pass. Elaborate multistage pipelines get replaced by **a single, end-to-end model**

Shallow learning transformed the input into one or two representation spaces – not expressive enough for most problems. So humans went to great lengths to hand-engineer good representations first.

Property 2 — scalability

- Highly amenable to **parallelisation on GPUs** and specialised hardware, so it rides Moore's law directly.
- Trained by iterating over **small batches**, which means dataset size is **no longer an upper bound**.
- The only real bottleneck is available parallel compute — and that barrier keeps moving.

Property 3 — versatility and reusability

> Trained models are also **repurposable**. That is the idea behind **foundation models**: very large models trained on enormous data, reusable across many new tasks with little retraining, or none.

SECTION 1.7

The cost of that power

> The property that makes it strong – **representations it discovers for itself** – is exactly the property that makes it hard to explain. Chapter 10 returns to this for image models.

In regulated settings, explainability is not an academic preference. It is a requirement, and it has to be designed for rather than retrofitted.

06

The age of generative AI

What changed in 2022, and what actually made it possible.

Self-supervised learning removed the labelling bottleneck

> So the targets are taken **from the input itself**. That is **self-supervised learning**, and it is what lets these models use vast amounts of *unlabelled* data.

Doing away with manual annotation – the bottleneck of every earlier generation of machine learning – unlocked a scale never seen before.

The scale that resulted

10^{11}
order of parameters in a foundation model

> 1 PB
order of training data

$10s M
order of cost to train one

2022
the year it entered public awareness

These models behave like a **fuzzy database of human knowledge**. Because they have already memorised so much, they solve new problems by **prompting** – no special-purpose programming, no retraining.

> It is newer in public attention than in research: the earliest text generation experiments date to the **1990s**, and the first edition of this book, in 2017, already had a chapter on generative models.

07

What has actually been achieved

The record, before the argument about the forecasts.

Three waves in a single decade

2013 - 2017
Perception
image, speech, handwriting

→

2017 - 2022
Language
translation, NLP, the Transformer

→

2022 - now
Generative
chat, code, image synthesis

Perception matured first, language second, generation third.

> Useful for planning: the capabilities that matured **earliest** are the cheapest and most reliable to deploy today. The newest ones are **the most expensive and the least settled**

Breakthroughs on problems that had long resisted machines

Conversation
ChatGPT, Gemini, Claude.

Code assistance
GitHub Copilot and its kin.

Photorealistic images
Generated from text.

Human-level perception
Image classification, speech transcription, handwriting.

Translation & speech
Both dramatically improved.

Autonomous driving
Deployed to the public in Phoenix, San Francisco, Los Angeles and Austin as of 2025.

Superhuman play
Go, chess, and poker.

Recommenders
YouTube, Netflix, Spotify.

And problems thought impossible a few years ago

Ancient manuscripts
Tens of thousands in the Vatican Secret Archive, transcribed automatically.

Plant disease
Detected and classified in the field with an ordinary smartphone.

Medical imaging
Assisting oncologists and radiologists with interpretation.

Natural disasters
Predicting floods, hurricanes, and even earthquakes.

This list is **not a forecast**. Every item on it is already running. That is what separates it from the next section.

08

Beware the short-term hype

Twice before, the field promised too much and lost its funding.

What was promised in 2023, and what happened

- Soon after GPT-4, pundits claimed **nobody would need to work** and that mass unemployment was a year away.
- Others promised economic productivity would rise **10**× **to 100**×.
- Two years later: US unemployment remains low, and productivity is far from any such explosion.

> To be fair to the technology: by mid-2025 generative AI was earning **tens of billions of dollars a year** – extremely impressive for an industry that **did not exist three years earlier**

SECTION 1.11

The gap that has to close somehow

> $200 bn

annual AI investment, mostly data centres and GPUs

≈ $30 bn

annual revenue generated against it

> *AI is currently being judged by executives and investors not by what it has accomplished, but by what we are told it might soon become able to do — much of which will durably stay out of reach of existing technologies. Something will have to give.*
>
> Chollet & Watson, section 1.11

SECTION 1.10

The order is the opposite of what people assume

It is tempting to think the practical success of generative AI produced the belief in near-term AGI. **It was the other way round**

1. **2013** — fears among tech elites that AGI was a few years out. The candidate was **DeepMind**, a London startup later acquired by Google.
2. **2015** — that belief drove the founding of **OpenAI**, intended as an open-source counterweight to DeepMind.
3. **2016** — OpenAI's recruiting pitch was that it would achieve **AGI by 2020**. Only a minority in the industry believed that timeline.
4. **Early 2023** — a significant fraction of Bay Area engineers were convinced AGI was one or two years away.

> OpenAI was critical in kick-starting generative AI. So in a peculiar twist, **the belief in near-term AGI fuelled the rise of generative AI**, not the reverse.

Winter, the first time

> *Within a generation . . . the problem of creating "artificial intelligence" will substantially be solved.*
>
> Marvin Minsky, 1967

> *In from three to eight years we will have a machine with the general intelligence of an average human being.*
>
> Marvin Minsky, 1970

When those expectations failed to materialise, researchers and government funding turned away, and the **first AI winter** began.

Winter, the second time — and where we are now

The 1980s: expert systems. A few early successes triggered a wave of investment; by around 1985 companies were spending **over $1 billion a year**.

By the early 1990s these systems had proven expensive to maintain, difficult to scale, and limited in scope. Interest died.

Now. Chollet's own view: a full-scale retreat like the 1990s is unlikely – AI has already demonstrated world-changing value. **If there is a winter, it should be very mild.** But some air will have to come out.

SECTION 1.10

Cognitive automation is not intelligence

> *Despite its name, today's "artificial intelligence" is more accurately described as "cognitive automation" — the encoding and operationalization of human skills and knowledge.*
>
> Chollet & Watson, section 1.10

It excels at problems with narrowly defined requirements, or where ample precise examples exist. It is about **enhancing the capabilities of computers**, not replicating human minds.

The cartoon and the living being

AI is like a cartoon character; intelligence is like a living being. A cartoon, however realistic, can only act out the scenes it was drawn for.

"If the cartoon is drawn with sufficient realism and covers sufficiently many scenes, what's the difference?"

> The difference is **adaptability**. Intelligence is the ability to face the unknown, adapt to it, and learn from it. Automation, even at its best, can only handle situations it has been trained on.

That is why building robust automation is so hard: it requires accounting for **every possible scenario**

What follows for anything you put into production

> If a system can only handle what it was trained on, then **monitoring for data drift is not an optional feature** – it is a condition of the system remaining fit to use. Chapter 6 and chapter 18 return to this.

And the reassuring corollary: do not worry about AI suddenly becoming self-aware. Today's technology is not headed that way. *"It's like expecting a better clock to lead to time travel — they're just different things altogether."*

09

The promise of AI

The short-term forecasts deflate. The long-term change still arrives.

SECTION 1.12

A 2017 forecast, scored in 2025

WRITTEN IN 2017	WHERE IT STANDS IN 2025
AI as your assistant, even your friend	**Tens of millions** use chatbots as daily assistants. Hundreds of thousands interact with AI "friends" in apps such as Character.ai.
It will answer your questions and help educate your kids	Question-answering and homework assistance turned out to be the **top two** applications.
It will drive you from point A to point B	Fully autonomous driving is deployed at scale in Phoenix, San Francisco, Los Angeles, and Austin.
It will help scientists make breakthroughs	**AlphaFold** predicts protein structures. Terence Tao expects AI to be a reliable co-author in mathematical research around 2026.

SECTION 1.12

Two things that are true at the same time

The short-term hype will deflate

The forecasts of 2023 have not held, and the investment-to-revenue gap has to close.

The long-term change still comes

Across medicine, science, industry, and daily life — the direction was right even where the timing was wrong.

> Holding both at once is uncomfortable, and it is **exactly what makes planning in this field difficult**

SUMMARY

What has to survive this chapter

1. **AI ⊃ machine learning ⊃ deep learning.** Symbolic AI is inside AI and outside machine learning.
2. **Machine learning inverts programming**: data and answers in, rules out.
3. **Three required ingredients** — inputs, example outputs, and a measure of quality. The third goes missing most often.
4. **"Deep" means many layers of representation**, not deeper understanding.
5. **Weights, loss, backpropagation** — the training loop chapter 2 takes apart.
6. **The achievements are real; the forecasts have repeatedly not been.** This is cognitive automation, not yet intelligence.

NOTEBOOK	NEXT	BOOK CODE
01_the_ml_paradigm.ipynb	Chapter 2 — Mathematical building blocks	fchollet/deep-learning-with-python-notebooks